Class-Responsibility-Collaborator (CRC)

To identify the main classes in the game and how they interact, we began our design with Class-Responsibility-Collaborator (CRC) cards. Each card names a class, lists its responsibilities (what it knows and does) and its collaborators (the other classes it works with). We chose CRC cards because they are quick to create and revise, which suited the early stage of the project, when the architecture was still developing. Walking through gameplay scenarios with the cards helped us identify missing responsibilities and unnecessary dependencies before any code was written, and they formed the basis of our initial class diagram.

We identified six core classes from the requirements:

- **Game**: manages the overall game state, timer and win/loss conditions (UR_GAME_END)
- **Player**: handles movement and the inventory (UR_MOVEMENT, UR_ITEMS)
- **Goose**: the obstacle that approaches in waves and blocks the player (UR_GEESE)
- **Item**: picked up and used by the player to disperse geese (UR_ITEMS)
- **Event**: triggerable occurrences that positively or negatively affect the player (UR_EVENTS)
- **Score**: tracks progress and rewards efficient item use (UR_SCORE)

The same class names are used consistently across our CRC cards, diagrams and code, so each class can be traced back to the requirements it satisfies. The CRC cards are available on our team website.

Game	
Responsibility	**Collaboration**
Show the 5 minute countdown	
Track the score of the player	
Store the state of the game (Paused/Playing)	
Tracks the number of items used	Items
(tracking player input (keypresses))	Player
Tracks the number of events triggered	Events
Track player not leaving map boundary	Player
Timer ends and player loses the game	Player

Events	
Responsibility	**collaboration**
Spawns when triggered	Game
Interacts with geese	Goose
Interacts with player	Player
Spawns additional objects (possibly)	Item
Affects game rules (possibly) (eg score % bonus)	Game
Play audio when triggered	Game

Goose	
Responsibility	**collaboration**
Spawns in waves depending on game countdown	Game
Pathfinds to player	Player
Knows its own position and state	
Dispersal of geese when triggered by item	Item
Responds to events	events
Leave when dispersed	Item
Status bar measuring geese appetite	Game, Player

Item	
Responsibility	**Collaboration**
Manage item spawning	Game
Store item properties	
Apply effects on geese when item used by player	Goose
Apply effects on player	Player
Starting with 5 items for the player	Player, Game

Player	
Responsibility	**Collaboration**
Stores and uses items	Item
Responds to events	Event
Item use triggered by users input	Item
Movement on map based on users input	
Health bar ? affected by geese	Game

Score	
Responsibility	**Collaboration**
Is it affected by Item usage (on geese?)	Item, Goose?
Score increases when triggered by goose dispersing	Goose
Affects whether the User wins or loses	Game
When 100% geese dispersed, game ends	Game